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資工一 B 線性代數期末考試題 2025/06/16

$$\begin{bmatrix} 4 & 10 \\ 3 & 0 \\ 0 & 20 \end{bmatrix}$$

I.

$$A = \begin{bmatrix} 4 & 0 & 0 & 1 \\ 1 & 2 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & 1 & 2 \\ -1 & 0 & 3 & 1 \end{bmatrix}$$

$$-1 \begin{bmatrix} 1 & -1 & -1 \\ 3 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$$

(i) 利用降階法(使用餘因子展開)計算 $\det(A)$ 。

(ii) 不用降階法與 2×2 及 3×3 之矩陣行列式公式計算 $\det(B)$ 。

$$\frac{1}{\sqrt{5}}$$

II. 求 $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ 之 QR- 分解。即求矩陣 Q 與 R , 使 $A = QR$, 其中 Q 之行向量成單範正交, R 為上三角矩陣。

$$0 - \frac{1}{2} \quad \frac{1}{2}$$

$$r_{11} = \sqrt{4} = 2$$

$$\frac{1}{2}$$

$$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$$

III. 設一數列 $\langle x_n \rangle_{n=1}^{\infty}$ 滿足

$$x_0 = 1, \quad x_1 = 2, \quad x_{n+2} = -x_{n+1} + 2x_n$$

$$x_{n+1} = -x_n + 2x_{n-1}$$

利用矩陣的對角化求 x_n 之一般式。

$$\begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$1, 2, 0, -1, -4, 12, -20, 44, -84, \dots \quad a_n = (-2)^{n+1}$$

IV. $T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} x_1 - x_2 \\ x_2 + x_3 \end{bmatrix}$. 求一矩陣 A 使 $[T(v)]_{B'} = A[v]_B$, $v \in \mathbb{R}^3$ 其中

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad B' = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

V.

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -2 & 5 & -2 \\ -6 & 6 & -3 \end{bmatrix}$$

將 A 對角化。即求一可逆矩陣 P 與一對角矩陣 D 使 $P^{-1}AP = D$ 。

$$1. \quad A = \begin{bmatrix} 4 & 0 & 0 & 1 \\ 1 & 2 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & 1 & 2 \\ -1 & 0 & 3 & 1 \end{bmatrix}$$

$$2. \quad 4 \begin{vmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 1 & -2 \end{vmatrix} - \begin{vmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 4(-4-0) - (-1-0) = -16+1 = -15$$

3.

$$\begin{vmatrix} 5 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & 1 & 2 \\ -1 & 0 & 3 & 1 \end{vmatrix} \xrightarrow{\substack{r_1=r_1+r_4 \\ r_2=r_2+r_4}} \begin{vmatrix} 4 & 1 & 3 & 2 \\ 0 & 0 & 2 & 0 \\ 2 & 0 & 1 & 2 \\ -1 & 0 & 3 & 1 \end{vmatrix} \xrightarrow{\substack{r_2=\frac{1}{2}r_2 \\ \text{swp}(r_2, r_3)}} \begin{vmatrix} 4 & 1 & 3 & 2 \\ 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 3 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 4 & 1 & 0 & 2 \\ 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{vmatrix} \xrightarrow{\substack{r_1=r_1-3r_3 \\ r_2=r_2-r_3 \\ r_4=r_4-3r_3}} \begin{vmatrix} 4 & 1 & 0 & 2 \\ 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{vmatrix} \xrightarrow{r_2=r_2+2r_4} \begin{vmatrix} 4 & 1 & 0 & 2 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{vmatrix} \xrightarrow{\substack{r_4=\frac{1}{4}r_4 \\ \text{swp}(r_2, r_4)}} \begin{vmatrix} 4 & 1 & 0 & 2 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

$$\xrightarrow{r_2=r_2-r_4} \begin{vmatrix} 4 & 1 & 0 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \xrightarrow{r_1=r_1-4r_2-2r_4} \begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \xrightarrow{\text{swp}(r_1, r_2)} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1 \quad \div \left(\frac{1}{2} \times \frac{1}{4} \right) = 8$$

II.

$$A = \begin{bmatrix} \cancel{x_1} & \cancel{x_2} & \cancel{x_3} \\ 1 & 0 & -1 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \text{QR 分解}$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$v_2 = x_2 - \text{Proj}_{v_1}(x_2) = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \end{bmatrix} - \frac{\vec{v}_1 \cdot \vec{x}_2}{\|\vec{v}_1\|^2} \vec{v}_1 = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \end{bmatrix} - \frac{4}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$$v_3 = x_3 - \text{Proj}_{v_1}(x_3) - \text{Proj}_{v_2}(x_3)$$

$$= \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \frac{\vec{v}_1 \cdot \vec{x}_3}{\|\vec{v}_1\|^2} \vec{v}_1 - \frac{\vec{v}_2 \cdot \vec{x}_3}{\|\vec{v}_2\|^2} \vec{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \end{bmatrix} - \begin{bmatrix} -\frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \\ -\frac{1}{4} \end{bmatrix} = \begin{bmatrix} -\frac{2}{4} \\ 0 \\ 0 \\ \frac{2}{4} \end{bmatrix}$$

標準化:

$$\frac{1}{\sqrt{4}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$\frac{1}{\sqrt{4}} \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$\frac{1}{\sqrt{\frac{1}{2}}} \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix} = \sqrt{2} \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ 0 \\ 0 \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{\sqrt{2}}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$R = Q^T A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{\sqrt{2}}{2} & 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2 & -\frac{1}{2} \\ 0 & 2 & \frac{1}{2} \\ 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix} \times$$

III、

$\langle x_n \rangle_1^\infty$ 為一數列, $x_0=1, x_1=2, x_{n+2}=-x_{n+1}+2x_n$
 $x_n=-x_{n-1}+2x_{n-2}$

$$\begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_{n-1} \\ x_{n-2} \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}^{n-2} \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix}$$

$$\det(\lambda I - A) = 0 \quad (\lambda I - A)E, P = [E_1, E_2, \dots, E_n], D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ \vdots & \lambda_2 & \dots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix}, A = PDP^{-1}$$

$$\begin{vmatrix} \lambda+1 & -2 \\ -1 & \lambda \end{vmatrix} = (\lambda^2 + \lambda) - 2 = (\lambda-1)(\lambda+2), \lambda = 1 \text{ or } -2$$

$\lambda = 1$:

$$\begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{aligned} 2x - 2y &= 0 \\ -x + y &= 0 \end{aligned}, E_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda = -2$

$$\begin{bmatrix} -1 & -2 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$-x - 2y = 0, E_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$$

$$\begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix} = PD^{n-2}P^{-1} \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1^{n-2} & 0 \\ 0 & -2^{n-2} \end{bmatrix} \times \frac{1}{3} \begin{bmatrix} -1 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= -\frac{1}{3} \begin{bmatrix} 1^{n-1} & 2 \times 2^{n-1} \\ \dots & \dots \end{bmatrix} \begin{bmatrix} -4 \\ -1 \end{bmatrix} = -\frac{1}{3} \begin{bmatrix} -4 \times 1^{n-1} + 2^n \end{bmatrix}$$

$$\Rightarrow x_n = \frac{4}{3} - \frac{1}{3}(-2)^n$$

$\mathcal{I}\mathcal{V}$.

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ x_2 + x_3 \end{bmatrix}, \text{ find a Matrix } A$$

$$[T(v)]_{B'} = A[v]_B, \quad v \in \mathbb{R}^3. \quad B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}, \quad B' = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$$

$$T\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 - 1 \\ 1 + 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$x + y = 0, \quad x = -y$$

$$x + 3y = 1, \quad 2y = 1, \quad y = \frac{1}{2}, \quad x = -\frac{1}{2}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} x+y \\ x+3y \end{bmatrix} \Rightarrow \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 - 1 \\ 1 + 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 2 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{cases} x+y = -1 \\ x+3y = 2 \end{cases} \Rightarrow \begin{array}{l} - \\ \hline -2y = -3 \end{array}$$

$$y = \frac{3}{2}, \quad x = -1 - \frac{3}{2} = -\frac{5}{2} \Rightarrow \begin{bmatrix} -\frac{5}{2} \\ \frac{3}{2} \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 - 0 \\ 0 + 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$A = \begin{bmatrix} -\frac{1}{2} & -\frac{5}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

N.

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -2 & 5 & -2 \\ -6 & 6 & -3 \end{bmatrix}, \text{ 对角化 } A.$$

(求一可逆矩阵P和对角矩阵D s.t. $P^{-1}AP = D$)

$$\lambda I - A = \begin{vmatrix} \lambda-1 & -2 & 2 \\ 2 & \lambda-5 & 2 \\ 6 & -6 & \lambda+3 \end{vmatrix}$$

$$(\lambda-1) \begin{vmatrix} \lambda-5 & 2 \\ -6 & \lambda+3 \end{vmatrix} + 2 \begin{vmatrix} 2 & 2 \\ 6 & \lambda+3 \end{vmatrix} + 2 \begin{vmatrix} 2 & \lambda-5 \\ 6 & -6 \end{vmatrix}$$

$$= (\lambda-1)(\lambda^2-2\lambda-3) + 2(2\lambda-6) + 2(-6\lambda+18)$$

$$= (\lambda^3 - 2\lambda^2 - 3\lambda - \lambda^2 + 2\lambda + 3) - 8\lambda + 24$$

$$= \lambda^3 - 3\lambda^2 - 9\lambda + 27$$

$$(\lambda-3)(\lambda^2-9)=0$$

$$= (\lambda-3)(\lambda-3)(\lambda+3)=0$$

$$= (\lambda-3)^2(\lambda+3)=0, \lambda=3 \text{ or } -3$$

重根

$\lambda=3$: (重根)

$$\begin{bmatrix} 2 & -2 & 2 \\ 2 & -2 & 2 \\ 6 & -6 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$2x-2y+2z=0$$

$$6x-6y+6z=0$$

$$E_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, E_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$\lambda=-3$:

$$\begin{bmatrix} -4 & -2 & 2 \\ 2 & -8 & 2 \\ 6 & -6 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$-4x-2y+2z=0$$

$$2x-8y+2z=0$$

$$6x-6y+0=0$$

$$E_3 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix}, D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -3 \end{bmatrix} \times$$

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(20分) 1. $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ 求 A 之 QR -分解。

2. 一科學試驗得以下數據：

(20分) $(x_1, y_1) = (0, 0), (x_2, y_2) = (2, 2), (x_3, y_3) = (3, 6), (x_4, y_4) = (4, 12)$.
求 x, y 之二次關係式 $y = f(x) = ax^2 + bx + c$ 在最小平方差的觀點下最符合以上實驗數據。(即求 a, b, c 之值使 $\sum_{i=1}^4 (f(x_i) - y_i)^2$ 最小)

(20分) 設 $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, u = \text{col}(A)$.

(5分) (a) 求一矩陣 P 使 對所有 $v \in \mathbb{R}^4, \text{Proj}_u(v) = Pv$.

(5分) (b) $v = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$, 求 v 在 $u = \text{col}(A)$ 上的投影向量。

(20分) 4. 計算 $\begin{bmatrix} 1 & 0 & 1 & 0 & 2 \\ -9 & 0 & 3 & 0 & 0 \\ 1 & 2 & 3 & 4 & 5 \\ 5 & 0 & 0 & 0 & 3 \\ 9 & 2 & 2 & 0 & 0 \end{bmatrix}$.

(20分) 5. $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 是否可對角化？若是，將其對角化，即求一可逆矩陣 C 與一
對角矩陣 Λ 使 $C^{-1}AC = \Lambda$ 。

(20分) 6. 一數列 $a_1, a_2, a_3, \dots, a_n, \dots$ 滿足 $a_{n+1} = 3a_n + 2a_{n-1}, n = 1, 2, 3, \dots, n, \dots$,

$a_0 = a_1 = 1$, 利用矩陣之對角化求 a_n 之般式。

(40分) 7. 設 $v_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, v_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \in \mathbb{R}^3, \mathcal{R} = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid \begin{bmatrix} x \\ y \\ z \end{bmatrix} = s \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + t \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}, s, t \in [0, 1] \right\}$

利用矩陣之QR分解之過程及行列式之知識

證明 $\mathcal{R}$ 之面積 $= \sqrt{\det(A^T A)}$, 其中 $A = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \\ z_1 & z_2 \end{bmatrix}$. (Hint: $A = QR, Q^T Q = I$)

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \text{QR 分解}$$

$x_1 \ x_2 \ x_3$

$$v_1 = x_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$v_2 = x_2 - \text{Proj}_{v_1}(x_2) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 - \frac{1}{2} \\ 1 - 0 \\ 1 - \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} \neq$$

$$v_3 = x_3 - \text{Proj}_{v_1}(x_3) - \text{Proj}_{v_2}(x_3) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}$$

標準化: $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

$$v_3 = \frac{3}{2\sqrt{3}} \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \end{bmatrix}$$

$$v_2 = \frac{\sqrt{6}}{3} \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{\sqrt{6}}{6} & \frac{1}{\sqrt{3}} \\ 0 & \frac{\sqrt{6}}{3} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{\sqrt{6}}{6} & -\frac{1}{\sqrt{3}} \end{bmatrix}$$

$$R = Q^T A = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{\sqrt{6}}{6} & \frac{\sqrt{6}}{3} & \frac{\sqrt{6}}{6} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{6}}{2} & \frac{\sqrt{6}}{6} \\ 0 & 0 & \frac{2\sqrt{3}}{3} \end{bmatrix} \neq$$

2,

(0,0), (2,2), (3,6), (4,12)

$$A^T A x = A^T y$$

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, A^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix}, y = \begin{bmatrix} 0 \\ 2 \\ 6 \\ 12 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix}, A^T y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 6 \\ 12 \end{bmatrix} = \begin{bmatrix} 20 \\ 70 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 20 \\ 70 \end{bmatrix} \quad \begin{array}{l} 4a + 9b = 20 \\ 9a + 29b = 70 \end{array} \Rightarrow \begin{array}{l} 36a + 81b = 180 \\ 36a + 116b = 280 \\ \hline -35b = -100 \quad b = \frac{20}{7} \end{array}$$

$$4a + \frac{9}{1} \times \frac{20}{7} = 20$$

$$4a = \frac{140}{7} - \frac{180}{7} = -\frac{40}{7}, a = -\frac{10}{7}$$

$$\begin{bmatrix} -\frac{10}{7} \\ \frac{20}{7} \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} \Rightarrow y = \frac{20}{7}x - \frac{10}{7}$$

4.

$$\begin{array}{c} \checkmark \\ \left| \begin{array}{cccc} 1 & 0 & 1 & -0 & 2 \\ -9 & 0 & 3 & +0 & 0 \\ 1 & 2 & 3 & -4 & 5 \\ 5 & 0 & 0 & +0 & 3 \\ 9 & 2 & 2 & -0 & 0 \end{array} \right| \end{array}$$

$$0 \left| \begin{array}{ccc} -9 & 0 & 3 & 0 \\ 1 & 2 & 3 & 5 \\ 5 & 0 & 0 & 3 \\ 9 & 2 & 2 & 0 \end{array} \right| + 0 \left| \begin{array}{ccc} 1 & 0 & 1 & 2 \\ 1 & 2 & 3 & 5 \\ 5 & 0 & 0 & 3 \\ 9 & 2 & 2 & 0 \end{array} \right| - 4 \left| \begin{array}{ccc} 1 & 0 & 1 & 2 \\ -9 & 0 & 3 & 0 \\ 5 & 0 & 0 & 3 \\ 9 & 2 & 2 & 0 \end{array} \right| + 0 \left| \begin{array}{ccc} 1 & 0 & 1 & 2 \\ -9 & 0 & 3 & 0 \\ 12 & 3 & 5 & \\ 9 & 2 & 2 & 0 \end{array} \right| - 0 \left| \begin{array}{ccc} 1 & 0 & 1 & 2 \\ -9 & 0 & 3 & 0 \\ 1 & 2 & 3 & 5 \\ 5 & 0 & 0 & 3 \end{array} \right|$$

$$= -4 \begin{array}{c} \checkmark \\ \left| \begin{array}{ccc} 1 & -0 & 1 & 2 \\ -9 & +0 & 3 & 0 \\ 5 & -0 & 0 & 3 \\ 9 & +2 & 2 & 0 \end{array} \right| \end{array}$$

$$= -4 \left(-0 \left| \begin{array}{cc} -9 & 3 & 0 \\ 5 & 0 & 3 \\ 9 & 2 & 0 \end{array} \right| + 0 \left| \begin{array}{cc} 1 & 1 & 2 \\ 5 & 0 & 3 \\ 9 & 2 & 0 \end{array} \right| - 0 \left| \begin{array}{cc} 1 & 1 & 2 \\ -9 & 3 & 0 \\ 5 & 0 & 3 \end{array} \right| + 2 \left| \begin{array}{cc} 1 & 1 & 2 \\ -9 & 3 & 0 \\ 5 & 0 & 3 \end{array} \right| \right)$$

$$= -4 \times 12 = -48 \#$$

$$\begin{array}{c} \left| \begin{array}{ccc} 1 & 1 & 2 \\ -9 & 3 & 0 \\ 5 & 0 & 3 \end{array} \right| \\ (9+0+0) - (3+0+(-27)) \\ = 6 \# \end{array}$$

5、 $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 是否能对角化? 若能则对角化。

(求一可逆矩阵 C 和对角矩阵 A 使 $C^{-1}AC = \Lambda$)

$$\begin{vmatrix} \lambda & -1 & 0 \\ -1 & \lambda & 0 \\ 0 & 0 & \lambda-1 \end{vmatrix} = [(\lambda^3 - \lambda^2) + 0 + 0] - [0 + 0 + (\lambda-1)]$$

$$= \lambda^3 - \lambda^2 - \lambda + 1$$

$$= \lambda^2(\lambda-1) - (\lambda-1) = (\lambda-1)(\lambda^2-1)$$

$\lambda=1$ $\lambda=\pm 1$
重根

$\lambda=1$

$$\begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$x-y=0$

$-x+y=0$

$E_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, E_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\lambda=-1$

$$\begin{bmatrix} -1 & -1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$-x-y=0$

$-2z=0$

$E_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

$C = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$C^{-1} = \frac{1}{\det(C)} Z^T \quad Z = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 2 & 0 \end{bmatrix}, \quad Z^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 2 \\ 1 & -1 & 0 \end{bmatrix} \quad \det(C) = \frac{1}{2} \quad C^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix}$

$z_{11} = \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} = 1 \quad z_{12} = -\begin{vmatrix} 1 & -1 \\ 0 & 0 \end{vmatrix} = 0 \quad z_{13} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$

$z_{21} = -\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1 \quad z_{22} = +\begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} = 0 \quad z_{23} = -\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = -1$

$z_{31} = \begin{vmatrix} 0 & 1 \\ 0 & -1 \end{vmatrix} = 0 \quad z_{32} = -\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = 2 \quad z_{33} = \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} = 0$

$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6、數列 $a_1, a_2, a_3, \dots, a_n$ ，滿足 $a_{n+1} = 3a_n + 2a_{n-1}$ ， $n=1, 2, 3, \dots, n$

$a_0 = a_1 = 1$ ，find a_n - 般式。

$$a_n = 3a_{n-1} + 2a_{n-2}$$

$$\begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a_{n-1} \\ a_{n-2} \end{bmatrix} = \begin{bmatrix} a_n \\ a_{n-1} \end{bmatrix} \quad \begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix} = P D^{n-2} P^{-1} x \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$$

$$\begin{vmatrix} \lambda - 3 & -2 \\ -1 & \lambda \end{vmatrix} = \lambda^2 - 3\lambda - 2 = 0 \quad \frac{3 \pm \sqrt{9+8}}{2} = \frac{3 \pm \sqrt{17}}{2}$$

$$\lambda = \frac{3 + \sqrt{17}}{2}$$

$$\lambda = \frac{3 - \sqrt{17}}{2}$$

$$\begin{bmatrix} \lambda - 3 & -2 \\ -1 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} \lambda - 3 & -2 \\ -1 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\lambda x - 3x - 2y = 0$$

$$E_2 = \begin{bmatrix} \frac{3 + \sqrt{17}}{2} \\ 1 \end{bmatrix}$$

$$-x + \lambda y = 0, \quad x = \lambda y$$

$$x = \frac{3 + \sqrt{17}}{2} y, \quad E_1 = \begin{bmatrix} \frac{3 + \sqrt{17}}{2} \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} \frac{3 + \sqrt{17}}{2} & \frac{3 - \sqrt{17}}{2} \\ 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} \frac{3 + \sqrt{17}}{2} & 0 \\ 0 & \frac{3 - \sqrt{17}}{2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{3 + \sqrt{17}}{2} & \frac{3 - \sqrt{17}}{2} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(\frac{3 + \sqrt{17}}{2}\right)^{n-2} / 0 \\ 0 / \left(\frac{3 - \sqrt{17}}{2}\right)^{n-2} \end{bmatrix} \frac{1}{\sqrt{17}} \begin{bmatrix} 1 - \frac{3 - \sqrt{17}}{2} \\ -1 - \frac{3 + \sqrt{17}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} P^{-1} = \frac{1}{\sqrt{17}} \begin{bmatrix} 1 & -\frac{3 - \sqrt{17}}{2} \\ -1 & \frac{3 + \sqrt{17}}{2} \end{bmatrix}$$

$$= \frac{1}{\sqrt{17}} \begin{bmatrix} \left(\frac{3 + \sqrt{17}}{2}\right)^{n-1} & \left(\frac{3 - \sqrt{17}}{2}\right)^{n-1} \\ \dots & \dots \end{bmatrix} \begin{bmatrix} 1 - \frac{3 - \sqrt{17}}{2} \\ -1 + \frac{3 + \sqrt{17}}{2} \end{bmatrix}$$

$$= \frac{1}{\sqrt{17}} \left[\left(\frac{3 + \sqrt{17}}{2}\right)^{n-1} - \frac{3 - \sqrt{17}}{2} \left(\frac{3 + \sqrt{17}}{2}\right)^{n-1} - \left(\frac{3 - \sqrt{17}}{2}\right)^{n-1} + \left(\frac{3 + \sqrt{17}}{2}\right) \left(\frac{3 - \sqrt{17}}{2}\right)^{n-1} \right] \times$$

QR 分解:

$$A = [x_1, x_2, x_3]$$

做 Gram-Schmidt:

$$v_1 = x_1, \quad v_2 = x_2 - \text{Proj}_{v_1}(x_2), \quad v_3 = x_3 - \text{Proj}_{v_1}(x_3) - \text{Proj}_{v_2}(x_3)$$

後 v_1, v_2, v_3 記得做標準化!

$$Q = [v_1, v_2, v_3], \quad R = Q^T A$$

行列式: 某行 $\times k$ 倍, $\det(A)$ 也會 $\times k$ 倍 (要除回去!)

兩列 or 兩行互換 $\Rightarrow \det(A) \times -1$

$$\text{二階反方陣公式: } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\text{三階公式: } A^{-1} = \frac{1}{\det(A)} C^T$$

凡是降階操作

記得要對正負對應表!

$$C = \begin{bmatrix} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ -\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & -\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} & -\begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} & \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}$$

特徵值、特徵向量 & 對角化

$$\det(\lambda I - A) = \det(A - \lambda I) = 0 \rightarrow \text{特徵值} \quad D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_n \end{bmatrix} \leftarrow \begin{matrix} \text{對角} \\ \text{矩陣} \end{matrix}$$

$$(\lambda I - A)E = 0 \rightarrow E \text{ 為特徵向量}$$

$$A = P D P^{-1}, \quad D = P^{-1} A P \leftarrow \text{對角化}$$

$$P = [E_1, E_2, \dots, E_n]$$

D, E, λ 順序一定要對!

遞迴一般式

$$\begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix} = P D^{n-2} P^{-1} \begin{bmatrix} x_1 \\ x_0 \end{bmatrix}$$

e.g. $x_n = 2x_{n-1} + 3x_{n-2}$, $x_1 = 0$, $x_2 = 2$

$$\rightarrow \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_{n-1} \\ x_{n-2} \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix}$$

A, 拿这个去找 λ, P, D

線性變換

三维
 $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ x_2 + x_3 \end{bmatrix}$, find a Matrix A

$$[T(v)]_{B'} = A[v]_B, \forall v \in \mathbb{R}^3. \quad B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}, B' = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \right\}$$

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1-1 \\ 1+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\begin{aligned} x+y &= 0, & x &= -y \\ x+3y &= 1, & 2y &= 1, y = \frac{1}{2}, x = -\frac{1}{2} \end{aligned}$$

構造線性組合, $\rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} x+y \\ x+3y \end{bmatrix} \Rightarrow \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$

解 x, y 之值

$$T\left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0-1 \\ 1+1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 2 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{cases} x+y = -1 \\ x+3y = 2 \\ -2y = -3 \end{cases} \quad y = \frac{3}{2}, x = -1 - \frac{3}{2} = -\frac{5}{2} \Rightarrow \begin{bmatrix} -\frac{5}{2} \\ \frac{3}{2} \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0-0 \\ 0+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

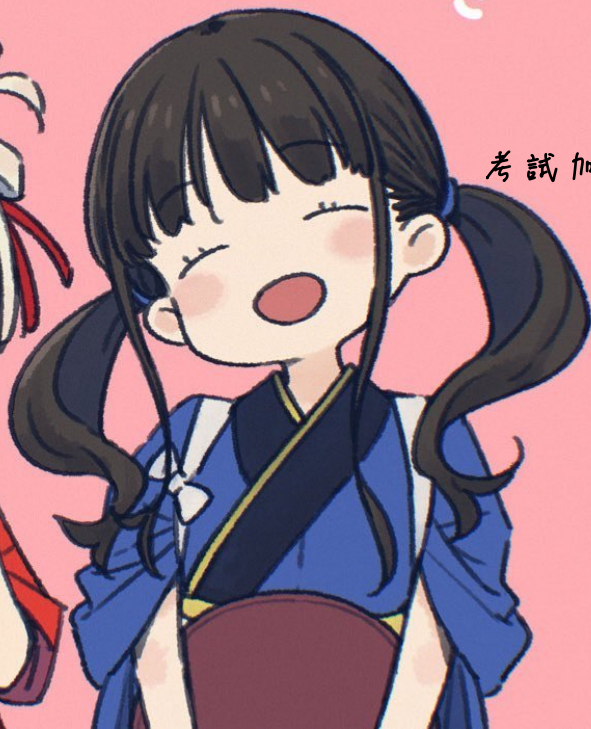
$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$A = \begin{bmatrix} -\frac{1}{2} & -\frac{5}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

リコショート

はじめました



考試加油!